AI-Powered Telecom Network Operations Assistant: architecture

Local execution · Python 3.12 · default LLM Claude Haiku 5.5 via the AICredits gateway · every LLM step has a deterministic fallback (zero-key mode)

OFFLINE BUILD · python -m scripts.bootstrap (no LLM)

Dataset
5G NetOps (HF fenar/5g_netops)
100,000 rows × 27 KPI columns
Apache-2.0 · 10k committed sample

Preprocessing
normalize · percentile baselines (p75/p90)
19 symptoms → 8 categories + NORMAL
priority P1–P4 · out-of-fold fault risk

Chunking
case narratives (1 per record,
abnormal KPIs only, template v1)
runbooks: 1 chunk per section

Embeddings
BAAI/bge-small-en-v1.5
384-d, CPU (sentence-transformers)
100k cases + 81 KB chunks

Vector Database
ChromaDB (persistent, local)
collections: cases (100,000), KB (81)
metadata: state, season, weather, category…

Models + baselines
joblib models · baselines.json
processed cases.parquet

ML training
fault-risk classifier, throughput regressor
leading indicators only (leakage audit)

KB runbooks
9 Markdown runbooks (authored,
reviewed; 3GPP / ITU-T refs)

BM25 index
bm25s keyword index
over the case narratives

dense + keyword indexes

ONLINE REQUEST · uvicorn src.main:app + npm run dev (frontend-react)

Retriever
hybrid: dense (Chroma) + BM25
reciprocal-rank fusion (k = 60)
rerank: symptom overlap + feedback prior
metadata filters, exact masked dense path
KB search by agent role

LLM/RAG Pipeline
evidence packs with citable IDs
LLM client: JSON + 1 repair, cache,
rate limit, circuit breaker,
per-run budget (12 calls / 50K tok),
daily spend cap

AICredits gateway
external · Claude Haiku 5.5, prepaid
Claude API / NVIDIA = config switch

Agents LangGraph state graph · judge-requested revision (max 1) · deterministic fallback per node

Classifier
category, symptoms,
filters, scope

Analyst
ranked root causes,
impact, citations

Recommender
ordered actions,
verification

Judge
scores + rule checks,
pass / revise / fail

Deterministic tools (no LLM)

KPI analyzer
bands, anomalies

Segment statistics
co-occurrence lifts

ML predictors
fault risk, throughput

Prioritization + impact
P1–P4, affected services

Similar-case search
via the Retriever

Guardrails
injection, scope, PII

Fallbacks: rules classifier · runbook root causes · runbook actions · rule-only validation

Reduced mode (no key / circuit open / spend cap): rules → retrieve → evidence → deterministic answer

models + baselines

API Layer
FastAPI /api/v1
13 endpoints
guardrails first
error envelope,
request IDs

User Interface
React (NetOps Triage)
Investigate
Cases
Simulate
Quality

tool call: similar cases / runbook sections

SQLite stores (artifacts/)
app.db: runs, ratings, per-case relevance votes
llm_cache.db: validated LLM replies (repeat queries are free)
llm_spend.db: estimated daily spend (cap LLM_DAILY_BUDGET_USD)

Beta-smoothed prior → rerank (weight 0.1)

feedback: ratings + case relevance votes (POST /feedback)

Legend
offline build step · deterministic (no LLM) · LLM component
storage / artifact · tool call / data read · feedback loop